TASK 05-50-27-200-801-A
EFFECTIVITY: ALL

1. Landing Gear Down Overspeed - Inspection/Check

A. General

(1) This task gives the procedure that you must do when at least one of the events below occurs:

- Retraction of the landing gear at an airspeed higher than the airspeed limit for landing gear operation.
- Extension of the landing gear at an airspeed higher than the airspeed limit for landing gear operation.
- Flight with the landing gear down and locked at an airspeed higher than the airspeed limit for flight with landing gear down.

(2) This maintenance technician is necessary to do this task:

- 1 - Mechanical Systems

(3) This maintenance technician is necessary to do this task:

- 1 - Electrical/Avionics Systems

B. References

REFERENCE	*DESIGNATION*
AMM TASK 32-00-00-840-801-A/200	Landing Gear - Prepare for Maintenance
AMM TASK 32-00-00-840-802-A/200	Landing Gear - Restore after Maintenance
AMM TASK 32-12-00-820-801-A/200	Main-Landing-Gear Doors - Adjustment
AMM TASK 32-12-01-860-801-A/200	Main-Landing-Gear Inboard Door (open) - Configuration
AMM TASK 32-12-01-860-802-A/200	Main-Landing-Gear Inboard Door (close) - Configuration
AMM TASK 32-22-01-820-801-A/200	Nose-Landing-Gear Forward-Door Rods Linkage - Adjustment
AMM TASK 32-22-02-860-801-A/200	Nose-Landing-Gear Forward Door (open) - Configuration
AMM TASK 32-22-02-860-802-A/200	Nose-Landing-Gear Forward Door (close) - Configuration
AMM TASK 32-22-03-820-801-A/200	Nose-Landing-Gear Aft-Door Rods Linkage - Adjustment
AMM TASK 32-30-01-710-801-A/500	Landing-Gear Extension-and-Retraction System (On Jacks) - Operational Test
AMM TASK 32-35-00-710-801-A/500	Emergency-Extension System - Free Fall (On Jacks) - Operational Test

C. Overspeed Landing Gear Operation - Event Assessment

SUBTASK 200-002-A

(1) The IAS limits for flight with landing gear in transit or locked down are given in section 2 of the AFM and named as follow:

- V_{LO} (e) is the maximum speed that the landing gear can be safely extended.
- V_{LO} (r) is the maximum speed that the landing gear can be safely retracted.
- V_{LE} is the maximum speed that the airplane can be safely flown with the landing gear down and locked.

(2) The maximum value speed is as follows to give the operational envelope for flight with the landing gear in transit or locked down:

(a) The V_{LO} (e) value is 250 KIAS.

(b) The V_{LO} (r) value is 220 KIAS.

(c) The V_{LE} value is 265 KIAS.

(3) The purpose of this section is to provide the operator with all information to analyze the flight data, determine the required inspections, and restore the aircraft airworthiness.

(4) For this event calculation, the parameters below must be selected from the flight data:

- Hours (GMT);
- Minutes (GMT);
- Seconds (GMT);
- Indicated air speed;
- Main landing gear position (Left or Right);
- Nose landing gear position.

D. Overspeed Landing Gear Operation - Exceedance Detection

SUBTASK 200-003-A

(1) Identify the flight period when the event occurred in the flight data.

(2) Examine the value of airspeed and landing gear control-lever position in the interval when the event occurred.

(3) Identify the samples where the airspeed (IAS) was higher than:

(a) The V_{LO} (e) value is 250 KIAS.

(b) The V_{LO} (r) value is 220 KIAS.

(c) The V_{LE} value is 265 KIAS.

(4) If speed value is higher than values above, do the SUBTASK 220-002-A.

E. Job Set-Up

SUBTASK 840-002-A

WARNING: **MAKE SURE THAT THE AIRCRAFT IS IN A SAFE CONDITION BEFORE YOU DO THE MAINTENANCE PROCEDURES. THIS IS TO PREVENT INJURY TO PERSONS AND/OR DAMAGE TO THE EQUIPMENT.**

(1) Do the procedure to make the aircraft safe for maintenance of the landing gear (AMM TASK 32-00-00-840-801-A/200).

SUBTASK 010-002-A

(2) Open the nose-landing-gear doors (AMM TASK 32-22-02-860-801-A/200).

SUBTASK 010-003-A

(3) Open the main-landing-gear inboard doors (AMM TASK 32-12-01-860-801-A/200).

F. Overspeed Landing Gear Operation - Inspection

SUBTASK 220-002-A

(1) Do these tasks to adjust the gaps and steps:

- AMM TASK 32-12-00-820-801-A/200;
- AMM TASK 32-22-01-820-801-A/200;
- AMM TASK 32-22-03-820-801-A/200.

(2) Do these operational tests:

- AMM TASK 32-30-01-710-801-A/500;
- AMM TASK 32-35-00-710-801-A/500.

(3) In case there are findings, send the flight data and the inspection results to EMBRAER and work together with EMBRAER for the necessary actions.

G. Job Close-Up

SUBTASK 940-002-A

(1) Remove from the work area all tools, equipment, and materials that you used.

SUBTASK 410-002-A

(2) Close the nose-landing-gear doors (AMM TASK 32-22-02-860-802-A/200).

SUBTASK 410-003-A

(3) Close the main-landing-gear inboard doors (AMM TASK 32-12-01-860-802-A/200).

SUBTASK 840-003-A

(4) Do the procedure to restore the aircraft after the maintenance of the landing gear (AMM TASK 32-00-00-840-802-A/200).